

Agenda

1. Exam 1 Regrade Requests are due today by 11:59 pm.
2. Midsemester Feedback

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3. Section 11.4

The Comparison Tests

Direct Comparison Test

Suppose that $0 \leq a_n \leq b_n$ for all $n \geq 1$.

1.) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

2.) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.

Recall: ① $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$
② $\sum_{n=1}^{\infty} ar^{n-1}$ converges if $|r| < 1$, diverges if $|r| \geq 1$

Example: $\sum_{n=1}^{\infty} \frac{1}{n^2+5}$

a.) What series does it make sense to compare this to?

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

b.) Does the series in part a) converge or diverge?

$p = 2 > 1$, series converges

c.) What inequality do we need to show to apply the direct comparison test? (Show it!)

$$0 \leq \frac{1}{n^2+5} \leq \frac{1}{n^2} \quad (n^2 \leq n^2+5)$$

d.) What can we conclude about the given series?

$\sum_{n=1}^{\infty} \frac{1}{n^2+5}$ converges by the direct comparison test
since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Activity: Determine if the series Converges or diverges

1.) $\sum_{n=1}^{\infty} \frac{e^{1/n}}{n}$ Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ (diverges $p=1$)

$$0 \leq \frac{1}{n} \leq \frac{e^{1/n}}{n} \quad 1 \leq e^{1/n}$$

Since $n \geq 1$, $e^0 \leq e^{1/n} \leq e^1 \rightarrow 1 \leq e^{1/n} \leq e$
 $\sum_{n=1}^{\infty} \frac{e^{1/n}}{n}$ diverges by the direct comparison test because $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

2.) $\sum_{m=1}^{\infty} \frac{2 + \sin^3(m+1)}{2^m + m}$ $\sum_{n=1}^{\infty} \frac{1}{2^n} \rightarrow 1 = 2 - 1 \leq 2 + \sin^2(m+1) \leq 2 + 1 = 3$

$\frac{2 + \sin^3(m+1)}{2^m + m} \leq \frac{3}{2^m}$ and $\sum_{n=1}^{\infty} 3\left(\frac{1}{2}\right)^m$ converges ($r = \frac{1}{2}$)

$\sum_{m=1}^{\infty} \frac{2 + \sin^3(m+1)}{2^m + m}$ converges by the direct comparison test
 because $\sum_{m=1}^{\infty} 3\left(\frac{1}{2}\right)^m$ converges

The Limit Comparison Test

What series does it make sense to compare $\sum_{n=1}^{\infty} \frac{n+1}{n^2+n+5}$ to? $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges $p=1$

Can we easily show this inequality?

No, because it is untrue. $0 \leq \frac{1}{n} \leq \frac{n+1}{n^2+n+5}$?

Limit Comparison Test

$$n^2+n+5 \leq n(n+1) = n^2+n$$

Suppose $a_n, b_n > 0$, for all $n \geq 1$ and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = C$, where $C \neq 0$

1.) If $\sum_{n=1}^{\infty} a_n$ converges, $\sum_{n=1}^{\infty} b_n$ converges

2.) If $\sum_{n=1}^{\infty} a_n$ diverges, $\sum_{n=1}^{\infty} b_n$ diverges

Example: Does $\sum_{n=1}^{\infty} \frac{n+1}{n^2+n+5}$ converge or diverge?

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{n}{n^2+n+5}} = \lim_{n \rightarrow \infty} \frac{n^2+n+5}{n^2+n} = 1 \neq 0$$

therefore $\sum_{n=1}^{\infty} \frac{n+1}{n^2+n+5}$ diverges by the

limit comparison test since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges

Activity: Determine whether the series converge or diverge

1.) $\sum_{k=1}^{\infty} \frac{k^{2/3} + 3\sqrt{k}}{2k^{3/3} + 1}$

2.) $\sum_{n=1}^{\infty} \frac{1}{10^n + \sin(n)}$

Activity: Determine if the series converges or diverges.

1.) $\sum_{n=5}^{\infty} \frac{6 + 8n + 9n^2}{3 + 2n + n^2}$

4.) $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

2.) $\sum_{n=1}^{\infty} \frac{3}{n^2 + 7n + 12}$

5.) $\sum_{n=1}^{\infty} \left(\frac{1}{n^2} + 1\right)^2$

3.) $\sum_{n=2}^{\infty} \frac{5^{n+1}}{7^{n-2}}$

6.) $\sum_{n=2}^{\infty} \frac{7}{n(n+1)}$